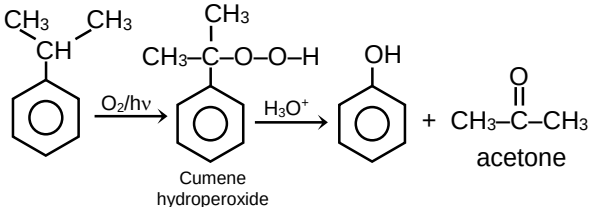
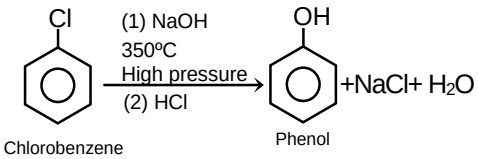
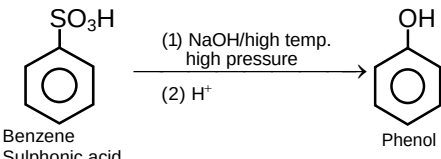
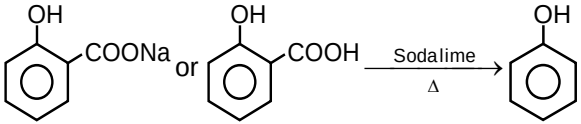
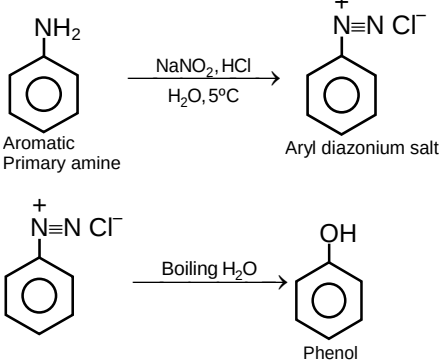




ABC-2 (Phenol & Aniline)

(A) PHENOL

Preparation of phenol (5-Methods)

1.	Cumene hydroperoxide method This is an industrial process to convert cumene into phenol and acetone in the presence of oxygen from air followed by hydrolysis.	 Cumene hydroperoxide
2.	Dow's process. In this process chlorobenzene is heated at 350°C (under high pressure) with sodium hydroxide which yields phenol.	 Chlorobenzene Phenol
3.	Fusion of benzene sulphonic acid with concentrated NaOH This is commercial process for synthesizing phenol. Benzene sulphonic acid is melted (fused) with sodium hydroxide at (300°C–320°C) followed by hydrolysis which yields phenol.	 Benzene Sulphonic acid Phenol
4.	Decarboxylation of salicylic acid Distillation of salicylic acid with soda-lime (NaOH + CaO) produces phenol.	
5.	Hydrolysis of benzene diazonium salt Benzene diazonium salt is prepared by reacting aromatic primary amine with NaNO ₂ , HCl at low temperature. Then this aqueous solution is heated to get phenol.	 Aromatic Primary amine Aryl diazonium salt Phenol





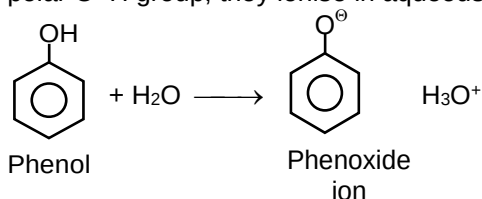
Chemical reactions of phenol [5-Reactions]

1	(a) Reaction with Br₂/H₂O When bromine is added to solution of phenol in presence of ethanol or H ₂ O, it forms white precipitate of 2,4,6-tribromophenol.	
	(b) Reaction with Br₂/CS₂ In presence of non-polar solvent (like CS ₂) or acids like CH ₃ COOH at low temperature, only monobromo product is obtained.	
2.	Reimer Tiemann formylation Phenol when heated with chloroform and NaOH followed by H ₂ O forms salicylaldehyde.	
3.	Reimer Tiemann carboxylation Phenol when heated with CCl ₄ and sodium hydroxide followed by hydrolysis forms salicylic acid. Note: Salicylic acid can be used in formation of aspirin. (Aspirin is used as painkiller)	
4.	Kolbe's Schmidt reaction Phenol when reacted with hydroxide ion in presence of CO ₂ forms a complex which on acidification forms salicylic acid.	
5.	Bakelite formation Bakelite is made by condensation reaction between phenol and formaldehyde.	

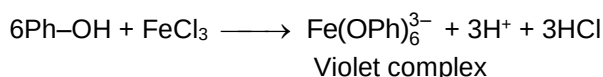


Lab test for phenol

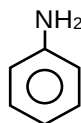
1. **Litmus test :** Phenol turns blue litmus red. Phenols behave as weak acid because of presence of polar O-H group, they ionise in aqueous solution to give H^+ ions.



2. **Neutral FeCl_3 :** Phenols give a violet-coloured water soluble complex with neutral ferric chloride.



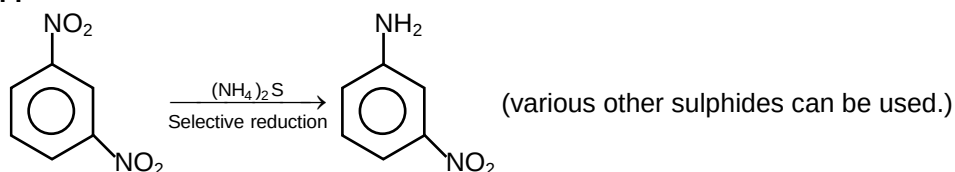
(B) ANILINE



Preparation of aniline [3-Methods]

1. Reduction by Metals	2. Reduction by H_2	3. Hofmann bromamide degradation reaction
		In this reaction an unsubstituted amides (only 1°) treated with NaOH/KOH and bromine to give a primary amine that has one carbon lesser than starting amide. $ \begin{array}{c} \text{O} \\ \\ \text{R}-\text{C}-\text{NH}_2 \end{array} + \text{Br}_2 + 4\text{NaOH} \longrightarrow \text{R}-\text{NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O} $ R can be : Alkyl or phenyl This method is used to prepare 1° aliphatic or aromatic amines.

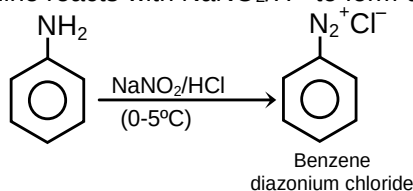
Special :



Chemical reactions of aniline :

1. Preparation of diazonium salt:

Aniline reacts with NaNO_2/H^+ to form diazonium salt.

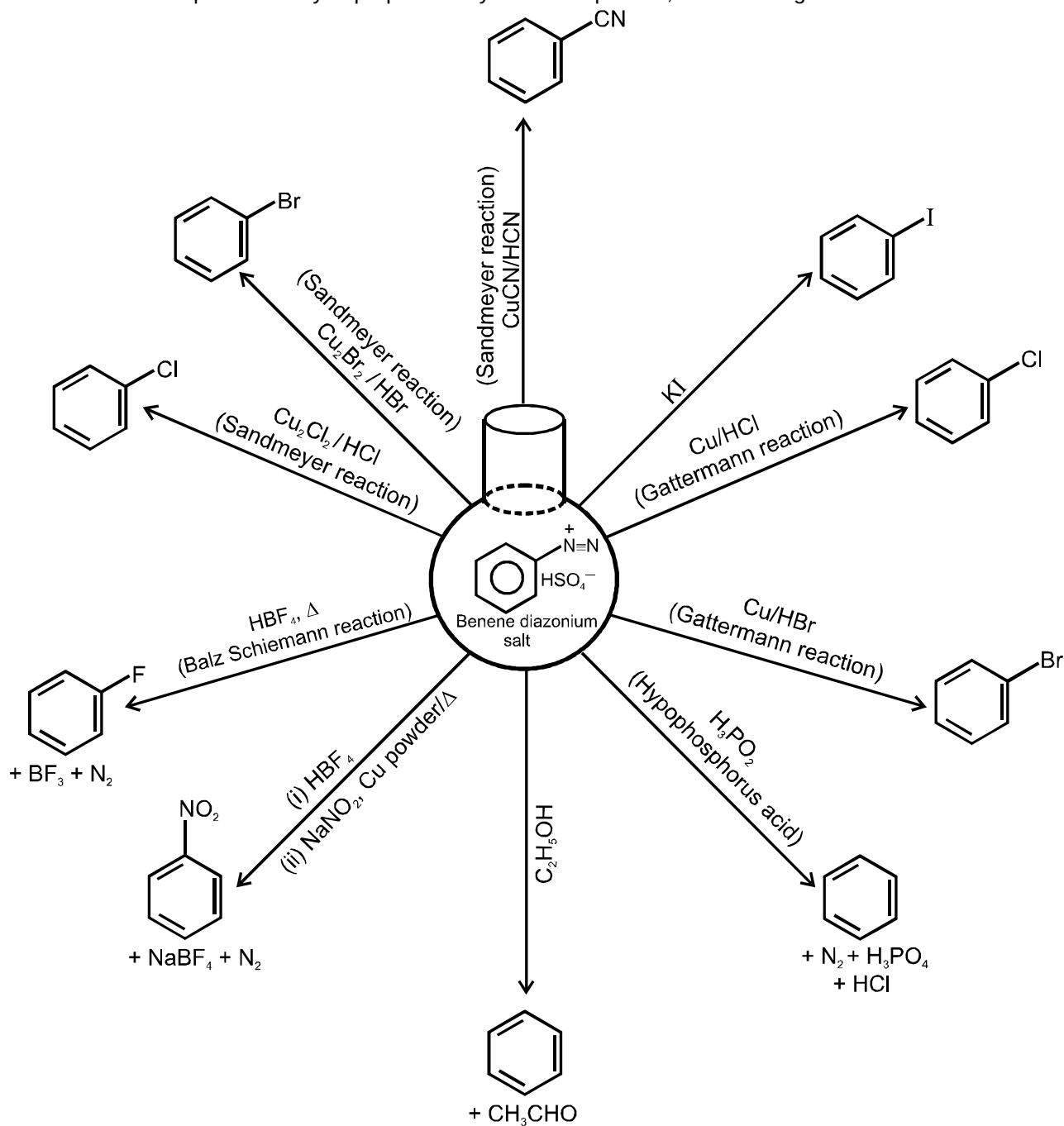


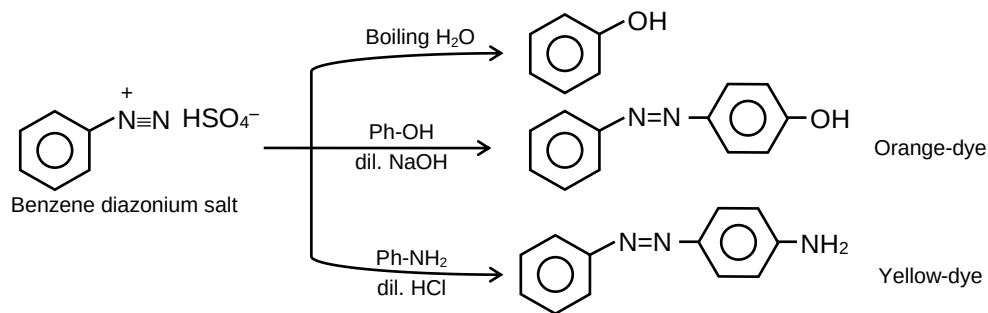
**Note :**

- Primary aliphatic amines** react with nitrous acid to form aliphatic diazonium salt which being unstable, liberate nitrogen gas.
- Primary aromatic amines form arene diazonium salts which are stable for a short time in solution at low temperatures (273-278 K). Due to its instability, the diazonium salt is not generally stored and is used immediately after its preparation.

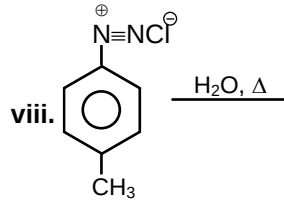
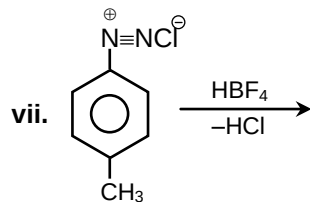
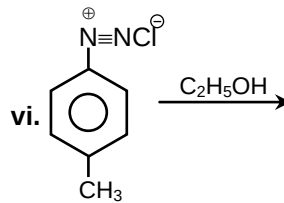
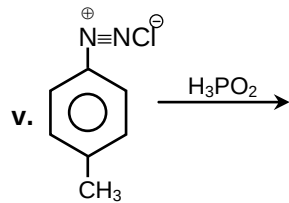
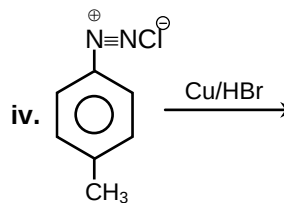
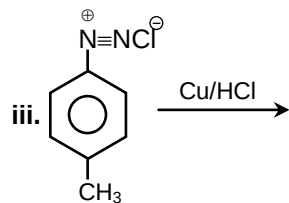
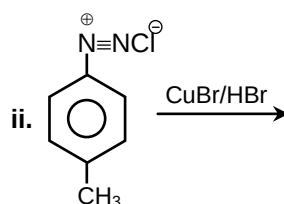
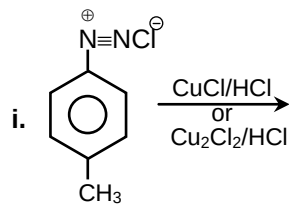
2. Chemical reactions of diazonium salt

Diazonium salt opens the way to prepare many other compounds, see following chart.

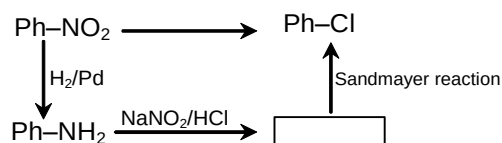




1. Complete the following reactions.



2. Conversion.

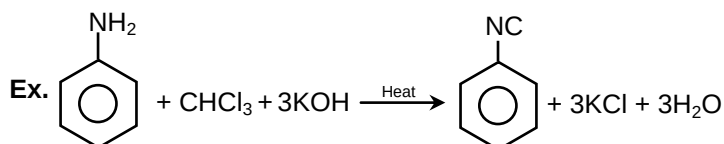
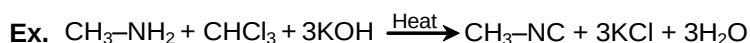
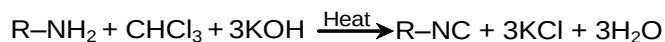




Lab test of aniline :

1. Carbylamine reaction

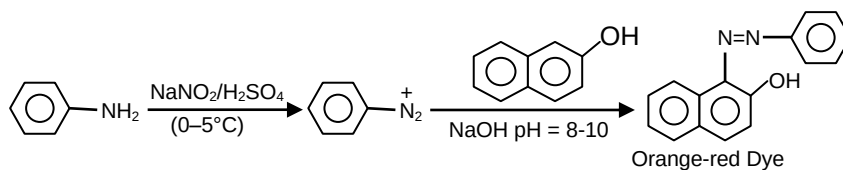
Primary amines (aliphatic as well as aromatic) react with chloroform (CHCl_3) on heating in the presence of ethanolic solution of KOH to form isocyanides (also called carbylamines) which are foul smelling substances. Secondary and tertiary amines do not undergo this reaction, therefore this reaction is used as a test for primary amines (aliphatic as well as aromatic).



2. Azo dye test

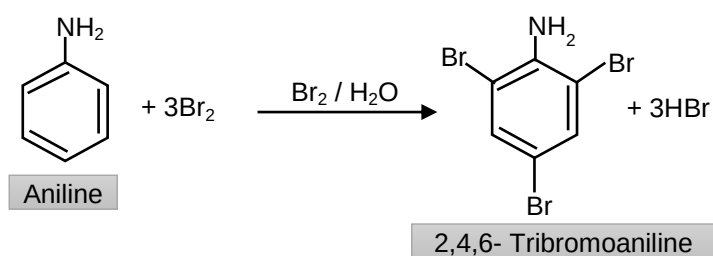
Azo compounds are usually intensely colored because of the azo linkage ($-\text{N}=\text{N}-$). Azo compounds, because of their intense color's and because they can be synthesized from relatively inexpensive compounds, are used extensively as dyes.

Synthesis of orange-red dye from 2-naphthol [β -naphthol] and aniline.



3. Bromine water test ($\text{Br}_2 + \text{H}_2\text{O}$):

Aniline reacts with bromine water at room temperature to give a white precipitate of 2,4,6-tribromoaniline. Aniline also gives test with $\text{Br}_2 + \text{CS}_2$



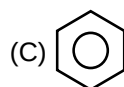
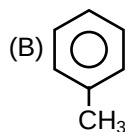
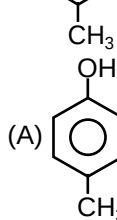
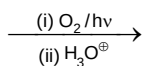
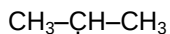


Exercise

ONLY ONE OPTION CORRECT TYPE

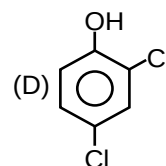
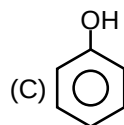
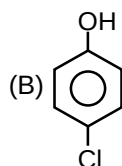
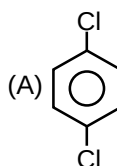
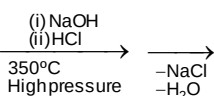
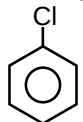
1. The process of conversion of cumene in the presence of oxygen and light followed by hydrolysis lead to the formation of _____.
 (A) Phenol (B) Aniline (C) Anisole (D) Benzene

2. Identify the product of following reaction.

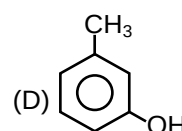
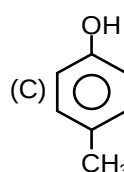
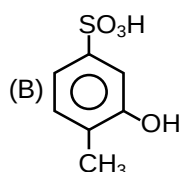
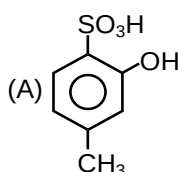


(D) None of these

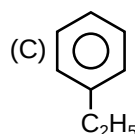
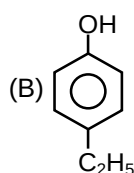
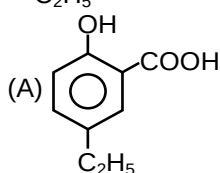
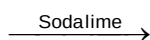
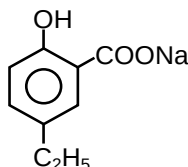
3. Identify the product of the following reaction.



4. $\xrightarrow[\text{(ii) H}^+]{\text{(i) NaOH}/\Delta}$?



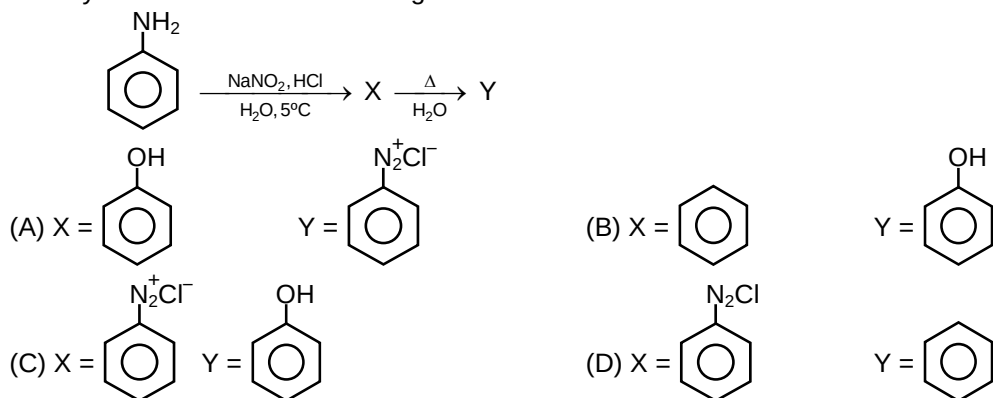
5. Give the product for following reaction.



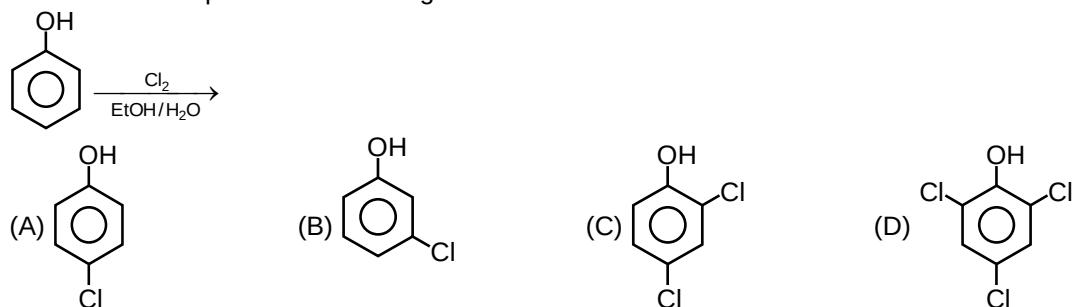
(D) None of these



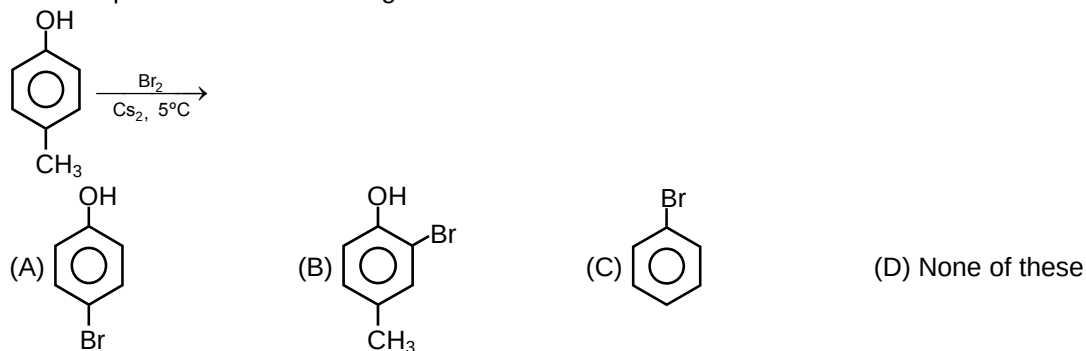
6. Identify the X and Y in the following reaction.



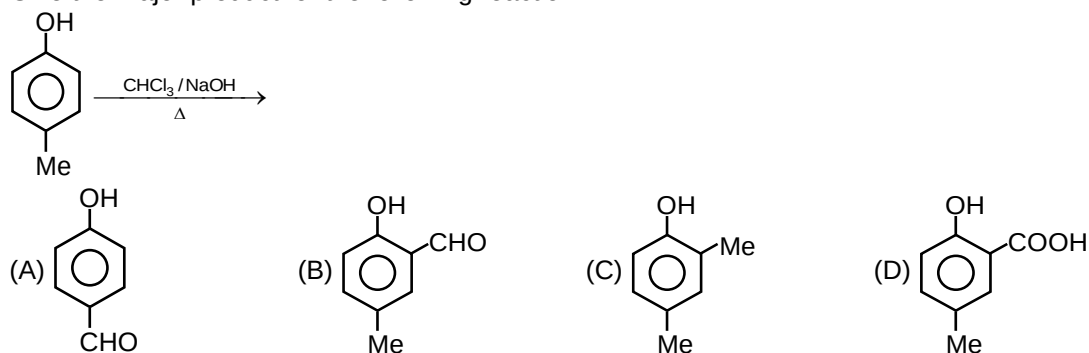
7. What will be the product of following reaction?



8. Find the product for the following reaction.



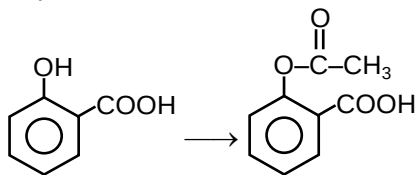
9. Give the major product for the following reaction.



10. The reagent used for Reimer Tiemann carboxylation in order to form salicylic acid is _____.
 (A) $\text{CHCl}_3 / \text{NaOH} / \text{H}^+$ (B) $\text{CCl}_4 / \text{NaOH} / \text{H}^+$ (C) $\text{OH}^- / \text{CO}_2 / \text{HCl}$ (D) None of these

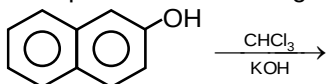


11. Aspirin can be formed from following reaction using which reagent?



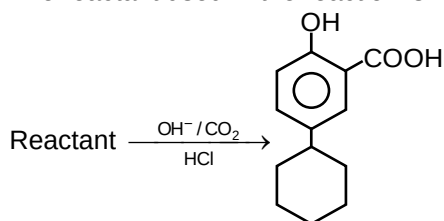
- (A) $(\text{CH}_3\text{--}\overset{\text{O}}{\parallel}\text{C--})_2\text{O} / \text{H}^+$ (B) $(\text{C}_2\text{H}_5\text{--}\overset{\text{O}}{\parallel}\text{C--})_2\text{O} / \text{H}^+$ (C) $\text{CHCl}_3 / \text{NaOH}$ (D) None of these

12. Give product for following reaction.



- (A) (B) (C) (D) None of these

13. The reactant used in the reaction is :



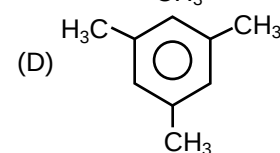
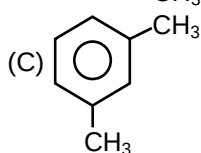
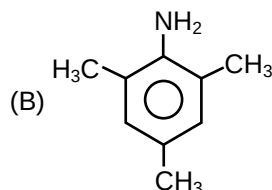
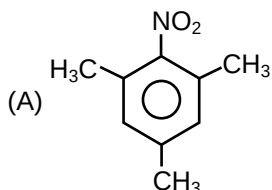
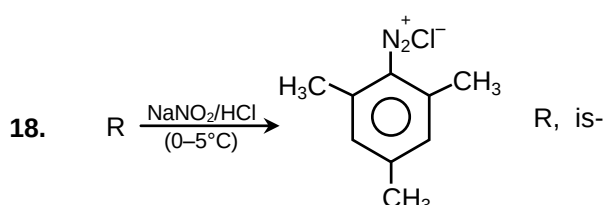
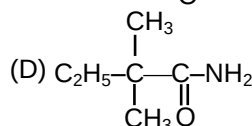
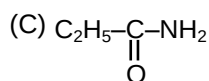
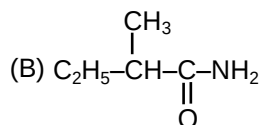
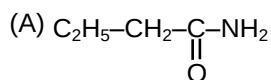
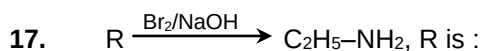
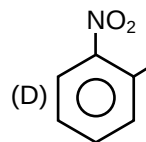
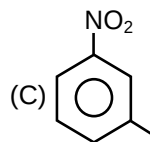
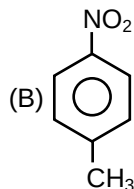
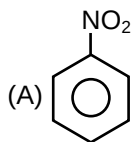
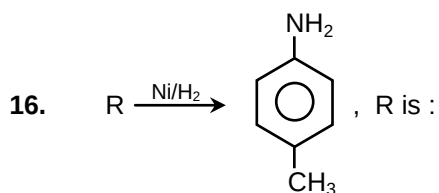
- (A) (B) (C) (D)

14. $\xrightarrow[+ 6\text{H}]{\text{Sn/HCl}}$ R, R is :

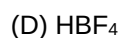
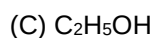
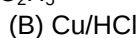
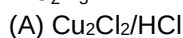
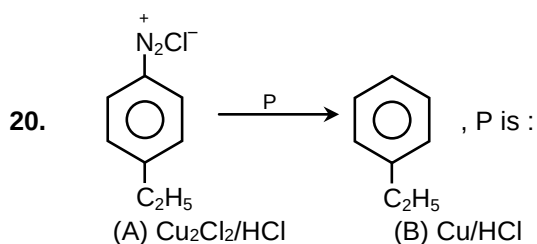
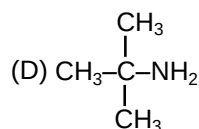
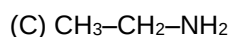
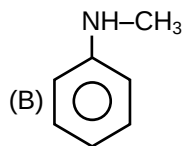
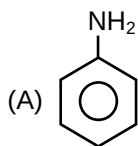
- (A) (B) (C) (D)

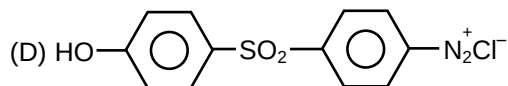
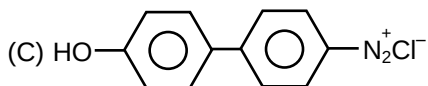
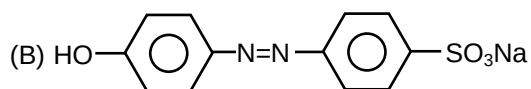
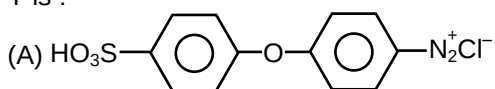
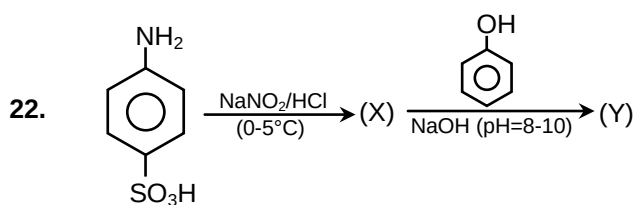
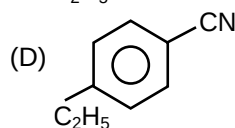
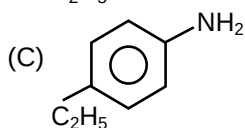
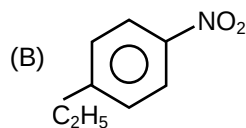
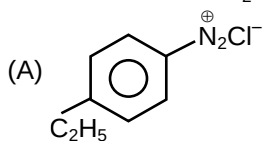
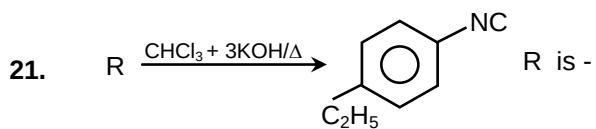
15. $\text{R} \xrightarrow[+ 6\text{H}]{\text{Fe/HCl}}$, R is :

- (A) (B) (C) (D)



19. Which of them produces diazonium cation at $0^\circ C$?





Answers

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (A) | 2. (A) | 3. (C) | 4. (C) | 5. (B) |
| 6. (C) | 7. (D) | 8. (B) | 9. (B) | 10. (B) |
| 11. (A) | 12. (B) | 13. (C) | 14. (C) | 15. (C) |
| 16. (B) | 17. (C) | 18. (B) | 19. (A) | 20. (C) |
| 21. (C) | 22. (B) | | | |